SHAAM PURUSHOTHAMAN JAYAKUMARAN

(236) 333-3792 jspurushothaman150904@gmail.com shaampjayakumaran jshaam7

EDUCATION & CERTIFICATIONS

The University of British Columbia *Vancouver, Canada*
B+MM Dual Degree Program: BSc Statistics + Master of Management *September 2022 - December 2028*

- **Dean's List Honouree (2024-2025)**
- **Coursework:** Statistical Modelling for Data Science, Finding Relationships in Data, Software Construction, Linear Programming, Stochastic Processes, Statistical Inference, Marketing, Accounting, Business Communications

Certifications: Google Data Analytics Professional Certification; International Certification in Digital Literacy

PROFESSIONAL EXPERIENCE

Ministry of Citizens' Services, BC Public Service Agency *Vancouver, Canada*
AI Analyst Co-op Intern, Technology Roadmap and Investment Planning *Sept 2026 - Present*

- Applied **System 1 models, Jev and Laya,** to entity resolution.
- Sourced real data to generate synthetic training data with **AUC 0.981 and Accuracy 0.924**.
- Researched in a containerized environment on **OpenShift**, supporting reproducible and privacy-aware experimentation.

UBC Data Science Institute *Vancouver, Canada*
Data Science Co-op Intern *May 2026 - Aug 2026*

- Analyzed **1.3M** household records using R-based survey fusion pipelines and **LightGBM** models to investigate relationships between consumption patterns and subjective well-being, supporting evidence-based policy research.
- Utilized **NLP** techniques to identify and match variables across multiple datasets, reducing manual review time by **400%**.
- Developed researcher-facing **Shiny** app carrying out specific-utility synthetic data validation for **fusionACS**.

Extern, Outamation *Remote*
AI & Data Engineering Extern *December 2025 - Present*

- Building a **RAG pipeline** with LlamaIndex and PyMuPDF that automatically extracts data from **200+** page mortgage documents, eliminating manual review and **speeding up processing**.
- Benchmarked open-source LLMs including **Mistral and Phi-2** using Python scripts and standard evaluation metrics, selecting a model that met target retrieval accuracy while staying within latency and inference-cost limits.

UBC Student Housing and Community Services *Vancouver, Canada*
Senior Residence Advisor - Totem Park *August 2024 – Present*

- Optimized well-being for **100+** students, applying **conflict-resolution**, ensuring a safe and supportive living environment.

PROJECTS

Cyclistic Bike-Share Company Strategy & Analysis *November 2025 - December 2025*

- Analyzed **800,000+** trip records using **R**, standardizing multi-year quarterly datasets to improve **data reliability**.
- Developed **Tableau** dashboard to analyze rider behavior across member and casual user segments.
- Delivered a data-driven **conversion strategy**, proposing business recommendations to target high-duration user segments.

Deepfake Detection CNN Model *February 2025 - March 2025*

- Built an end-to-end image classification pipeline for deepfake detection using **OpenCV** and **TensorFlow CNN**.
- Achieved **82% test accuracy**, and analyzed training behavior using learning curves, confusion matrix, and classification report to identify error trade-offs.

Hotel Management Software Application *January 2024 - April 2024*

- Developed a full-stack desktop application in **Java** using Object-Oriented Programming **(OOP)** and Design Patterns to automate guest lifecycles, room inventory, and billing workflows.
- Architected a data persistence layer using **JSON** and validated system reliability through **JUnit testing**.

Business Analytics and Consulting - BOLT Datathon Finalist *September 2023*

- Analyzed a dataset of **16,000+ unique profiles** using **Python** to identify service-utilization gaps across five age categories.
- Proposed a **cost-effective digital strategy** projected to **double social media impressions** from **60K to 120K**.

LEADERSHIP & VOLUNTEER EXPERIENCE

President - UBC AMS Data Science Club, **Agricultural Researcher - UBC Agrobot Design Team**

TECHNICAL SKILLS

Languages & Frameworks: Python, R, SQL, Red Hat OpenShift, Java, C#, HTML, Bash, RAG, NLP, LLM
Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, Tesseract, Seaborn
Tools: Tableau, Power BI, Microsoft Word, Excel, PowerPoint, Git, GitHub, VS Code, LaTeX, Figma BigQuery